

21 Alternating currents

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

An alternating voltage changes sign periodically. For a sinusoid $V_{\text{rms}} = V_{\text{peak}}/\sqrt{2}$; resistive power uses RMS current and voltage. An ideal transformer obeys $V_s/V_p = N_s/N_p$ and conserves input/output power. Transmission at high voltage lowers current and I^2R losses for given power.

Worked example: follow the reasoning

A sinusoidal supply with peak 325 V has RMS voltage ≈ 230 V. A $100\ \Omega$ heater dissipates $P = V_{\text{rms}}^2/R \approx 529$ W.

Quick check · multiple choice

A Which is RMS voltage of a sinusoid with peak voltage V_0 ? [1]

- A $V_0/\sqrt{2}$
- B $\sqrt{2}V_0$
- C $V_0/2$
- D $2V_0$

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 An ideal transformer has 1000 primary turns, 50 secondary turns and 240 V primary RMS. Determine secondary voltage and primary current if secondary current is 2.0 A. [3]

Final answer: _____

Answers and explanation

Quick check: A. $V_0/\sqrt{2}$

Question 1 [3 marks]

$V_s = 240 \times 50/1000 = 12 \text{ V}$ [1]; equal powers: $240 I_p = 12 \times 2.0$ [1], $I_p = 0.10 \text{ A}$ [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Physics 9702, 2025–2027. <https://www.cambridgeinternational.org/Images/664565-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.